



IILM UNIVERSITY

School of Computer Science and Engineering

SOFTWARE ENGINEERING & ARCHITECTURE

INTERNSHIP PRESENTATION

STUDENT

Vineet Tiwari

ENROLLMENT NO.

2410031061

PROGRAM

B.Tech CSE — 5th Semester

UNIVERSITY

IILM University, Greater Noida

ORGANIZATION

YuvaIntern

ROLE

Lead Software Engineer

Internship Overview

ORGANIZATION

YuvaIntern

DURATION

4 Weeks

ROLE

Lead Software
Engineer

DOMAIN

Software Engineering
& Architecture

PROJECT

Scalable E-Commerce
Platform

OVERALL RESPONSIBILITIES

- Designed a scalable microservices architecture for a proposed e-commerce platform
- Converted the architecture into a practical implementation plan with timelines and risk management
- Carried out a representative code review and quality assurance exercise
- Defined a performance testing and optimization strategy for high-traffic scenarios

WHY THIS PROJECT

E-commerce systems combine many challenges of high-traffic applications — concurrent users, interdependent services, and the need for strong security and reliability — making them an effective learning vehicle for software architecture and engineering practice.

About YuvaIntern



*An initiative of Henry Harvin Education,
working with NSDC and industry partners.*

- A virtual internship and job-simulation platform that bridges theoretical education and real-world professional responsibilities
- Follows a “learning by doing” approach, aligning simulations with actual job descriptions and responsibilities
- Offers opportunities across technology, finance, marketing, HR, design, and other sectors
- Technology tracks include software development, data analytics, and technical support
- Simulations are developed with industry experts; internships can be completed remotely at the learner's own pace

INTERNSHIP COMPLETION CERTIFICATE



CERTIFICATE OF EXPERIENCE

To Whomsoever It May Concern

This is to certify that **Vineet Tiwari** successfully completed an internship with **YuvaIntern** in the role of **Lead Software Engineer** **hyderabad, in.**

Throughout the tenure of this internship, Vineet demonstrated consistent dedication, professionalism, and a strong willingness to learn and grow. Their contributions were valued by the entire team at YuvaIntern, and they conducted themselves with integrity and commitment throughout the program.

We commend Vineet's hard work and wish them continued success in all future endeavors.

Certificate No: YI/2026/184183/409162

Date of Issue: September 01, 2026

FOR AND ON BEHALF OF YUVAINTERN



Kounal Gupta

Founder, [YuvaIntern.com](https://yuvaintern.com)

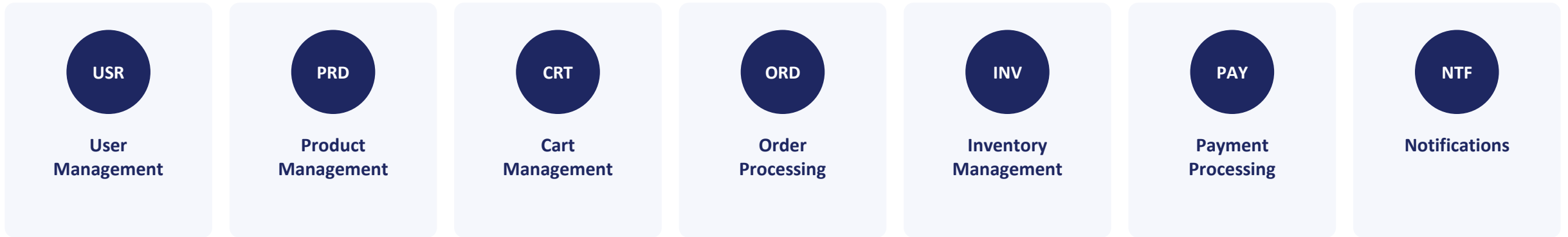
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Scalable E-Commerce Platform

A hypothetical, high-traffic e-commerce platform designed around a Microservices Architecture to explore scalability, maintainability, and independent service management.

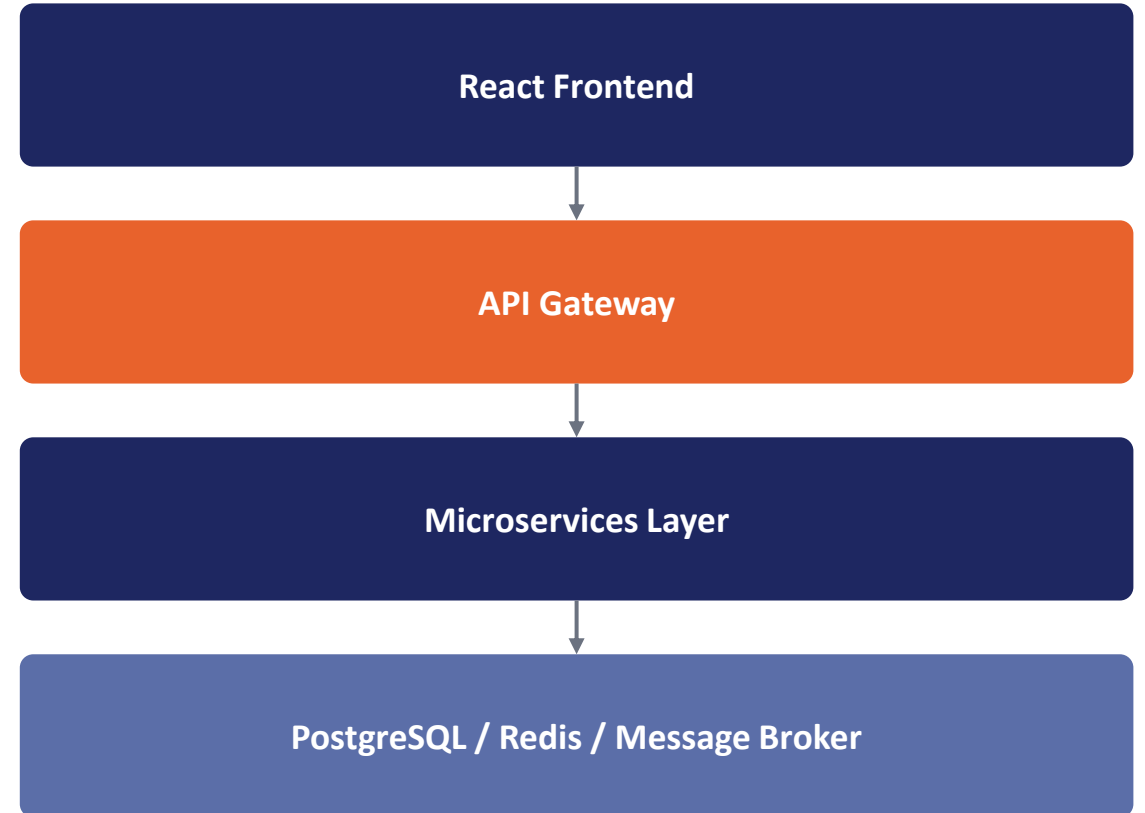


WHY MICROSERVICES

Dividing the platform into independent, loosely coupled services allows each business capability to be scaled, deployed, and maintained on its own — avoiding the single point of failure and tight coupling typical of monolithic designs.

Scalable Software Architecture

- Designed a microservices-based architecture for the platform
- Decomposed the system into independent business-capability services
- Introduced an API Gateway as the single client entry point
- Proposed PostgreSQL for persistent data and Redis for caching
- Considered a message broker for asynchronous communication
- Addressed security: authentication, authorization, input validation
- Focused on scalability and long-term maintainability



Simplified request flow through the proposed architecture

Project Planning



MAJOR RISKS → MITIGATION

Risk	Mitigation
Requirement changes mid-project	Iterative planning with clear milestone checkpoints
Integration issues between services	Well-defined API contracts and early integration testing
Security vulnerabilities	Security review built into the planning and QA phases
Delays / resource limitations	Realistic timeline with buffer and phased delivery

Code Review & Quality Assurance

Representative / hypothetical code-review exercise conducted against the proposed microservice designs — not a production codebase.

Identified Issue	Impact	Improvement
Sensitive payment data handled directly by the application	High	Use secure payment gateways / tokenization
Authorization not consistently enforced	High	Apply role-based, resource-level authorization
Insufficient input validation at API boundaries	High	Validate all requests at each service boundary
Secrets stored in source / configuration	High	Use environment variables / secret management
Repeated database reads without caching	Medium	Introduce Redis caching for high-read data
No clear timeout / retry policy	Medium	Add controlled timeouts, retries, resilience

Performance Testing & Optimization

TESTING TYPES

- Baseline
- Load
- Stress
- Spike
- Endurance
- Scalability

METRICS MONITORED

- Response time
- Throughput
- Error rate
- CPU / memory usage
- Database performance
- Cache & queue health

OPTIMIZATION APPROACHES

- DB / query optimization
- Redis caching
- Horizontal scaling
- Connection-pool tuning
- Async processing
- Resilience improvements

Technologies & Tools

BACKEND / APIS

- Spring Boot
- REST APIs

DATABASE / STORAGE

- PostgreSQL
- Redis

FRONTEND

- React

TESTING / DEVELOPMENT

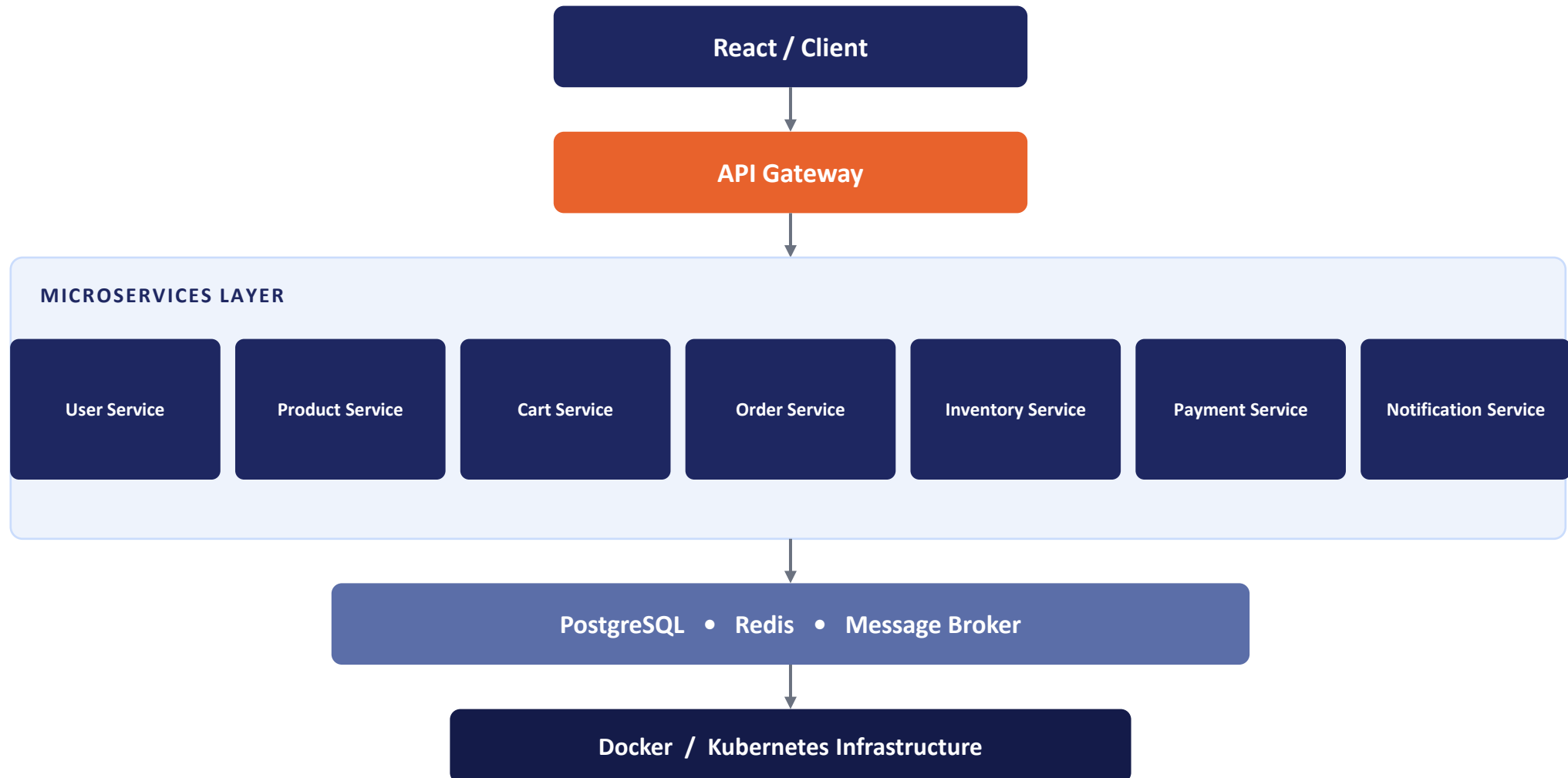
- Git
- k6 / JMeter

INFRASTRUCTURE / ARCHITECTURE

- Microservices
- Docker
- Kubernetes
- Message Broker
- API Gateway

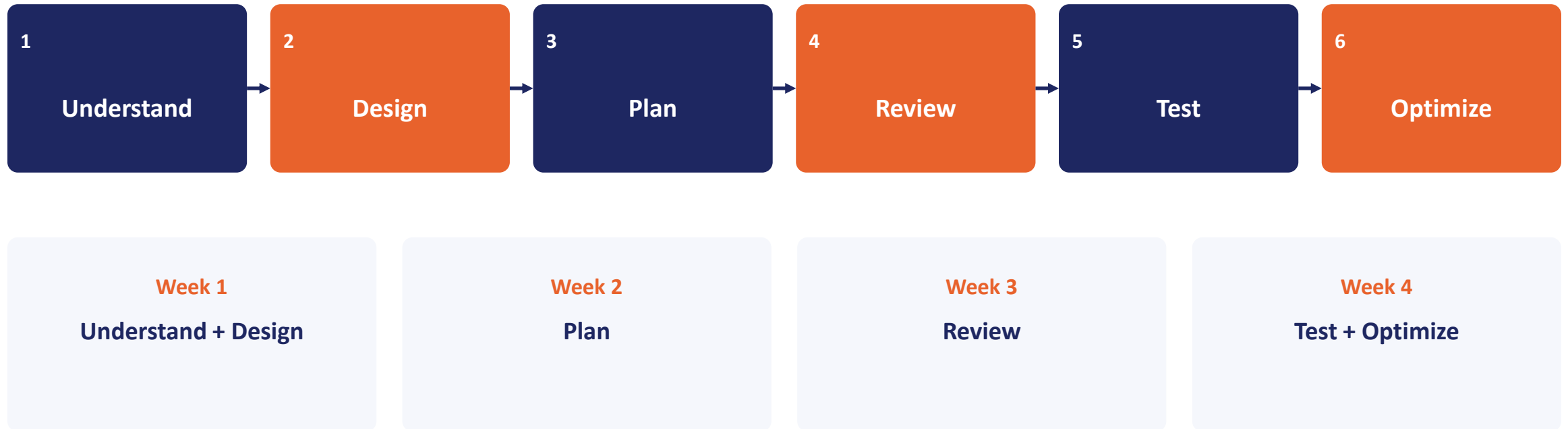
Technologies were used, proposed, or considered at the design level as part of the architectural exercise — not all were implemented in a running system.

System Architecture



Internship Methodology

A single coherent engineering progression — not four unrelated tasks.



Key Learning Outcomes

ARC

Scalable software architecture & microservices design

PLN

Practical project planning, task breakdown & risk management

SEC

Secure coding practices for auth & payment handling

REV

Code review and quality-assurance processes

PRF

Performance-testing types and evaluation metrics

OPT

Bottleneck identification & optimization strategies

SCL

Scalability and reliability in distributed systems

SDL

End-to-end view of the software engineering lifecycle

Role & Overall Contribution

LEAD SOFTWARE ENGINEER

Software Engineering & Architecture Internship — YuvaIntern

Contributions reflect the scope of the internship tasks; the platform itself remained a design-stage, project-based exercise.

- Led the design of the microservices architecture, including service decomposition and technology choices
- Drove technical planning — task breakdown, timelines, and risk management for the implementation plan
- Directed the representative code-quality review across the proposed services
- Defined the quality-assurance approach and key test areas
- Designed the performance-testing strategy and identified likely optimization approaches
- Owned key technical decisions connecting architecture, planning, quality, and performance

INTERNSHIP CREDENTIALS

Offer Letter — YuvaIntern



Dear Vineet Tiwari,

We are pleased to extend to you this formal offer for an Internship position as **Lead Software Engineer hyderabad**, in at **YuvaIntern**. We were impressed with your background, skills, and enthusiasm, and we believe you will be a valuable addition to our team.

Internship Details:

- **Start Date:** August 04, 2026
- **End Date:** September 01, 2026
- **Duration:** 4 weeks
- **Location:** Remote

Key Responsibilities:

As a **Lead Software Engineer hyderabad**, in intern, you will work on tasks and projects related to Internship program for Lead Software Engineer hyderabad, in..

Benefits & Learning Opportunities:

Throughout the duration of your internship, you will have the opportunity to:

- Develop practical skills in [relevant skills/tools].
- Gain hands on experience in [specific tasks/projects].

As part of joining our team, you may be required to sign a confidentiality or non-disclosure agreement. You are also expected to comply with all applicable company policies and regulations, which will be provided to you upon commencement.

A handwritten signature in black ink, appearing to read "Kounal Gupta".

Kounal Gupta
Founder, YuvaIntern

References & Thank You

REFERENCES

- YuvaIntern – About YuvaIntern (yuvaintern.com/about-us)
- Spring Boot Documentation (docs.spring.io/spring-boot)
- PostgreSQL Documentation (postgresql.org/docs)
- Redis Documentation (redis.io/docs)
- Docker Documentation (docs.docker.com)
- Kubernetes Documentation (kubernetes.io/docs)
- React Documentation (react.dev)
- k6 Documentation (grafana.com/docs/k6)

THANK YOU

Vineet Tiwari

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